



AI & Partners

Amsterdam - London - Singapore



EU AI Act

Modifying AI:

Best Practice for Compliance

January 2026

Authors

Sean Musch, Michael Borrelli, Charles Kerrigan, Stuart
Macdonald, Andrea Frosinini, Stella Tsitsoula, Kinga Incze



Regulation vs.



Measured Oversight

Classifying AI risks and ensuring compliance at every stage.



Transparent Practices

Clear documentation, audit trails, and traceable decision-making.



Safety First

Human oversight, ethical safeguards, and robust data protection.



Innovation



Adaptive Intelligence

Continuous learning, retraining, and model refinement.



Creative Experiments

Rapid testing, feature trials, and iterative improvements.



Performance Edge

Optimized efficiency, robustness, and scalable deployment.

AI now stands at the core of dual-use innovation

Dual Use Examples (December 2025)



-	Civilian Application	Defence Application
Materials	Carbon-fibre sheets for wind-turbine blades	Carbon-fibre for drone airframes
Hardware	High-powered lasers used in fibre optic comms	High-powered lasers for range finding/targeting
Software	Encryption for online banking and messaging	Encryption modules for secure comms and data
Technical Knowledge/Services	GPS integration know-how to build civilian sat-nav devices	GPS signal/anti-jamming know-how for military navigation

Their defining trait is purpose-neutral functionality: the same core features (e.g. sensing, positioning, computation, secure communications, control) can produce public-interest results in civilian environments or support security/defence roles in other contexts.

AI can be characterised as a General Purpose Technology (GPT). A GPT is defined by three key characteristics that ensure that they will have a large and positive impact on productivity and growth. The coincidence of (1) Rapid improvement, (2) Pervasiveness, and (3) Complementary innovations threatens to create profound change through the economy and society. As a GPT, AI has become a vital technology in the dual-use field. Although it is not typically classified as dual-use in control lists, it serves as a crucial dual-use enabler that transforms shared building blocks (data, models, compute) into civilian and security outcomes, ranging from autonomy and cyber security to decision support and critical infrastructure resilience.



AI & Partners

Amsterdam - London - Singapore

Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

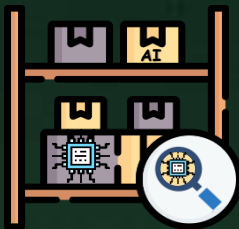
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

Contact: Team | contact@ai-and-partners.com

Contents

Who Are We	4
Introduction	6
Key questions being asked about Modifying AI under the EU AI Act	8
Issues with Modifications to AI under the EU AI Act	14
When are you qualified as a provider for AI modifications under the EU AI Act.....	17
General-Purpose AI in Action: A Practitioner’s Challenge	28
Going Forward.....	38
Conclusion.....	41
References.....	46

Introduction

High-Level Overview of the Subject

The European Union Artificial Intelligence Act (EU AI Act) represents one of the first comprehensive legislative frameworks aimed at regulating the development, deployment, and modification of artificial intelligence (AI) technologies within the European Union. As AI becomes increasingly integrated into daily operations across industries, from financial services and healthcare to enterprise software and public services, the governance of AI systems has emerged as a critical area of attention for regulators, businesses, and technical practitioners alike.

A key aspect of this governance is the regulation of modifications to AI systems and general-purpose AI (GPAI) models. Modifications can take many forms, including adjusting operational parameters, fine-tuning models with new data, or integrating models into broader applications. For example, a company providing customer service chatbots might adapt an existing large language model (LLM) by incorporating proprietary knowledge from internal databases. Similarly, a financial services firm could modify an AI-based credit scoring system to accommodate regional regulatory standards or organizational risk policies. In both cases, the act of modifying an AI system introduces questions regarding compliance responsibilities under the EU AI Act.

The significance of this subject extends beyond legal compliance. Modifying AI systems carries operational and ethical implications, influencing model behavior, systemic risk, and user outcomes. Proper understanding of the regulatory framework ensures that organizations can innovate and customize AI applications while mitigating potential legal, ethical, and operational risks.

Explanation of Key Terms and Concepts

Understanding the regulation of AI modifications under the EU AI Act requires defining several key concepts. First, the term **AI system** broadly refers to any software, model, or platform that is capable of producing outputs such as content, predictions, recommendations, or decisions that influence the environment in which it operates. These outputs may be used to support human decision-making, automate processes, or interact directly with end-users. A **general-purpose AI (GPAI) model** is a specific category of AI designed to perform across a wide range of tasks and use cases rather than being narrowly specialized for a single function. Examples include advanced language models or multimodal AI systems capable of processing both text and images. GPAI models are deliberately built to be adaptable and reusable, which makes them particularly relevant when considering regulatory obligations around modifications.

The term **modifier** refers to an individual or organization that alters an existing AI system or GPAI model, either by updating data inputs, changing hyperparameters, adjusting outputs, or integrating the model into a new system. Under the EU AI Act, the modifier may assume the role of a **provider**—the entity responsible for ensuring compliance with the Act—if the modification is deemed substantial. A **substantial modification** is generally understood as a change that affects the system's compliance with regulatory requirements, its intended purpose, or its overall risk profile. Such modifications may involve fine-tuning models with significant new data, distillation to a smaller architecture, or structural changes to the model's design. By contrast, minor modifications, such as adjusting

prompts or adding lightweight retrieval mechanisms, are typically considered insubstantial and may not trigger provider obligations. These concepts—AI system, GPAI model, modifier, provider, and substantial modification—interact in a framework where the degree and nature of a modification determine compliance responsibilities. The distinctions are essential for organizations seeking to understand whether their activities expose them to additional legal and operational obligations under the EU AI Act.

Conceptual Framework and System Description

The regulation of AI modifications under the EU AI Act can be conceptualized as a decision framework in which entities assess both the scope and impact of their actions on AI systems. At a high level, this framework considers:

1. **Original AI System or Model:** The base model or system as provided by the initial developer or upstream provider.
2. **Modification Activities:** Any change introduced by a downstream actor, ranging from integration of the AI system into a new application to fine-tuning or structural alterations.
3. **Assessment of Substantiality:** Evaluation of whether the modifications substantially affect the AI system's capabilities, generality, or risk profile.
4. **Shift of Provider Obligations:** Determination of whether the modifier becomes a provider under the EU AI Act, thereby assuming responsibility for compliance, documentation, and risk mitigation.

Analogous to maintaining a vehicle, this framework can be compared to assessing modifications to a car: routine maintenance like oil changes or tire rotations generally does not alter the vehicle's regulatory status. However, modifying the engine or safety systems substantially changes its operational

characteristics and could trigger additional compliance requirements under traffic or safety laws. Similarly, not all AI modifications are equivalent; only substantial modifications affect the regulatory responsibilities of the modifying entity.

Broader Context and Scope

The regulation of AI modifications under the EU AI Act exists within a broader landscape of organizational, operational, and societal considerations. On an organizational level, businesses integrating AI must consider not only compliance but also operational continuity, ethical implications, and reputational risk. Modifications to AI systems can influence outcomes such as decision accuracy, fairness, and transparency, which in turn affect stakeholder trust and regulatory scrutiny.

From a societal perspective, the regulation seeks to balance innovation with risk mitigation, ensuring that AI applications do not create undue harm, discrimination, or systemic instability. This includes distinguishing between AI systems that pose **high-risk** implications—such as those used in healthcare, transportation, or critical infrastructure—and lower-risk applications. In practice, organizations modifying AI systems must navigate these distinctions while implementing internal governance structures that support accountability, documentation, and risk assessment.

Applications of AI modification vary widely. In enterprise software, modifications often involve integration and configuration without altering the core AI model. In specialized domains, such as autonomous systems or financial risk modeling, modifications may involve extensive retraining or fine-tuning, potentially triggering compliance obligations. The EU AI Act establishes a scalable framework designed to capture these variations while providing clarity for downstream actors and modifiers.

Purpose and Scope of the Document

The purpose of this report is to provide a conceptual understanding and analytical overview of the regulatory considerations associated with modifying AI systems and GPAI models under the EU AI Act. It aims to clarify key concepts, outline practical considerations for organizations, and describe the decision-making framework for determining provider responsibilities. The report focuses specifically on the regulatory implications of **modifications to AI systems**, particularly those involving GPAI models, while deliberately excluding exhaustive technical guidance on AI system development or unrelated compliance regimes. It emphasizes the assessment of substantial modifications, risk profiles, and provider obligations, rather than prescribing operational methods or advocating specific implementation strategies. By presenting this information, the report seeks to equip organizations, technical practitioners, and policy analysts with a structured understanding of how modifications influence compliance responsibilities, supporting informed decision-making and proactive risk management. Readers can expect to gain clarity on:

- The distinction between minor and substantial modifications.
- The roles of modifiers and providers under the EU AI Act.
- Practical considerations for documentation, risk assessment, and governance.
- The broader context in which AI modifications are regulated and managed.

Overall, this introduction establishes the conceptual foundation for a detailed examination of AI modifications under the EU AI Act, supporting both analytical insight and practical comprehension. The subsequent sections of the report build on this foundation by exploring regulatory thresholds, practitioner case studies, and recommended approaches to navigating compliance obligations in a structured and systematic manner.



Key questions
being
asked about
Modifying AI
under the EU AI
Act



1. What constitutes a “modification” of an AI system under the EU AI Act?

A modification refers to any change made to an existing AI system or GPAI model that could affect its functionality, output, or compliance with regulatory requirements. Examples include fine-tuning on new datasets, adjusting hyperparameters, integrating new modules, or distilling the model into a smaller architecture. Modifications can be minor, such as changing prompts or adding retrieval-augmented generation (RAG), or substantial, such as retraining a model on significant proprietary data.

2. Who is considered a “provider” when modifying AI systems?

Under the EU AI Act, a provider is any entity that develops, deploys, or substantially modifies an AI system. When a downstream actor substantially alters a model or AI system—such as through fine-tuning or architectural changes—they may assume the role of a provider, taking on compliance responsibilities including documentation, risk assessment, and conformity obligations. Minor modifications typically do not trigger provider status. Determining whether a modifier becomes a provider depends on assessing the modification’s impact on the system’s risk profile, generality, and intended purpose, aligned with the Act’s definitions and guidance from the GPAI guidelines.

3. What is a “substantial modification” in the context of GPAI models?

A substantial modification occurs when changes to a GPAI model significantly alter its generality, capabilities, or systemic risk. Examples include fine-tuning on large, proprietary datasets, knowledge distillation to a smaller model, or structural changes affecting the model’s architecture. Minor adjustments like prompt changes, RAG

integration, or branding customizations usually do not qualify. The EU AI Act provides indicative thresholds, such as using at least one-third of the compute (FLOPs) required to train the original model, but emphasizes assessing actual changes in risk and behavior rather than computational effort alone.

4. How does the EU AI Act address risk in modifications?

Risk assessment is central to determining compliance obligations. Modifications are considered substantial if they change the AI system’s high-risk classification, affect safety, fairness, or reliability, or introduce systemic risks. For GPAI models, risk is evaluated in terms of potential societal impact, misuse, or unintended behaviors. Organizations modifying AI systems must conduct impact assessments to evaluate these risks, document findings, and adjust governance accordingly. Even when computational thresholds are not met, modifications that meaningfully alter outputs or performance require careful consideration under the Act.

5. What types of AI modifications usually do not trigger provider obligations?

In general, modifications that do not meaningfully alter the AI system’s purpose, risk profile, or capabilities are considered insubstantial. Examples include adjusting hyperparameters, creating prompt-based variants (Custom GPTs), adding lightweight retrieval or external knowledge sources, or integrating the model into broader applications without retraining. Such changes typically leave the original provider responsible for compliance. Nonetheless, organizations are encouraged to document all modifications, even minor ones, as best practice for governance and traceability.

6. What is the significance of the compute-based threshold in the GPAI guidelines?

The GPAI guidelines introduce a threshold using computational resources: if a modification consumes at least one-third of the compute (FLOPs) used to train the original model, the modifier is presumed to assume provider responsibilities. This threshold helps clarify when substantial modifications might occur, particularly in fine-tuning scenarios. However, it is indicative rather than absolute. Low-compute modifications can still materially affect risk, generality, or systemic behavior.

7. How should organizations assess whether a modification is substantial?

Assessment requires evaluating both the **scope** and **impact** of the modification. Key considerations include: whether the modification alters the system's intended purpose, generality, or capabilities; whether it affects risk profiles or systemic behavior; and whether it meets or exceeds indicative compute thresholds. Organizations should conduct a structured risk and impact assessment, documenting technical changes, intended effects, and residual risks. Cross-functional collaboration between legal, compliance, and technical teams is recommended to interpret guidelines. Maintaining this documentation supports governance, internal audits, and potential regulatory engagement.

8. What are typical examples of modifications that trigger provider obligations?

Substantial modifications typically include: fine-tuning a GPAI model with large proprietary datasets; distilling a model into a smaller architecture that changes output behavior; jailbreaking or reprogramming model parameters to override original safety constraints; or structural changes affecting generality or systemic risk. In these cases, the downstream actor may inherit provider obligations, including documentation, transparency measures, risk mitigation, and

compliance reporting. Minor adaptations, such as prompt tuning or integrating external knowledge bases, generally do not trigger obligations. Each case requires careful analysis to determine whether regulatory responsibilities shift under the EU AI Act.

9. How can organizations document modifications effectively?

Documentation should capture the **nature, purpose, and impact** of modifications, including technical specifications, datasets used, retraining processes, and any effect on risk or system behavior. Records should also note the decision-making rationale and any mitigation measures implemented. Maintaining structured documentation enables internal governance, risk management, and regulatory compliance, particularly during periods of legal uncertainty. Even when modifications are minor, documentation provides traceability and demonstrates due diligence.

10. What practical steps should businesses take when modifying AI under the EU AI Act?

Businesses should first classify the AI system or GPAI model and determine its high-risk status. Next, they should evaluate whether planned modifications are substantial by assessing changes to purpose, capabilities, and risk. Conducting risk and impact assessments, documenting modifications, and maintaining internal governance frameworks are critical. Organizations may adopt conservative approaches, treating any modification as potentially triggering provider obligations, or pragmatic approaches focused on demonstrable changes. Staying informed of EU AI Act updates, leveraging vendor documentation, and engaging with regulatory guidance or service desks supports compliance while enabling responsible innovation with AI systems

Technical Implementation of Modification Thresholds

While the EU AI Act frames “substantial modification” primarily in terms of impact on risk, generality, and intended purpose, organizations still require practical methods to assess when these thresholds may be met. From a technical perspective, this often involves combining computational measurement with structured change tracking.

One commonly referenced indicator is computational effort, particularly for fine-tuning or retraining general-purpose AI models. Organizations can estimate this by tracking floating point operations (FLOPs) consumed during modification activities and comparing them to the compute used in the original training run, where such information is available. While FLOPs-based thresholds are indicative rather than determinative, logging compute usage provides an objective signal that a modification may warrant closer regulatory scrutiny.

In parallel, organizations should implement version-controlled model management practices. This includes maintaining repositories that log model versions, datasets used, parameter changes, training scripts, and configuration files. Each modification should be accompanied by metadata describing its purpose, scope, and expected effect on system behaviour. Where models are deployed through MLOps pipelines, automated logging of training runs, evaluations, and deployments can further enhance traceability.

Importantly, technical measurement should not be treated in isolation. FLOPs tracking and version control support—but do not replace—qualitative assessment of behavioural change and risk impact. Together, these practices enable organizations to evidence diligence, support internal decision-making, and demonstrate a defensible approach to assessing modification thresholds under the EU AI Act.



**Andrea
Frosinini,**
*Business Development
Manager, XDC
Foundation*



Issues with Modifications to AI under the EU AI Act



The regulation of artificial intelligence (AI) within the European Union has introduced significant attention to how AI systems and models are developed, deployed, and modified. Among the most challenging areas of compliance arises when AI systems or models are altered by downstream actors after initial deployment. Such modifications can range from minor adjustments to substantial retraining or restructuring of models. Understanding the conceptual, analytical, and practical issues associated with AI modifications is critical for organizations, regulators, and technical practitioners seeking to operate responsibly and effectively within regulated ecosystems.

Conceptual Ambiguity and Definition Challenges

One of the primary issues in regulating AI modifications stems from conceptual ambiguity in defining what constitutes a “substantial modification” versus an insubstantial one. Substantial modifications are generally understood to be changes that materially affect the behavior, capabilities, or risk profile of an AI system. However, the boundary between substantial and minor changes is not always clear. Even small adjustments to parameters, data inputs, or operational configurations can have unforeseen consequences on outputs, particularly in complex, adaptive systems such as machine learning models.

This ambiguity creates challenges for organizations attempting to determine whether a regulatory obligation is triggered. For instance, an organization may adjust a model’s configuration to improve efficiency or integrate new data sources. While technically minor, these changes could produce outputs that deviate from original expectations, raising questions regarding responsibility for accuracy, fairness, or systemic impact. The lack of precise,

universally applicable criteria necessitates careful, case-specific evaluation and may lead to inconsistent interpretations across organizations and ecosystems.

Risk Assessment Complexity

Modifying AI systems inherently involves changes to the risk profile of the system. Assessing these risks requires understanding both direct and indirect effects of modifications. Direct effects may include changes in output accuracy, reliability, or the likelihood of system failure. Indirect effects, often more challenging to quantify, can involve emergent behavior, unintended biases, or interactions with other systems and users.

The complexity of risk assessment is exacerbated by the adaptive nature of many AI systems, particularly models that learn continuously or respond dynamically to new inputs. Even minor modifications can propagate through a system in ways that are difficult to predict, leading to risk exposures that were not present in the original design. Consequently, organizations must adopt structured risk evaluation methodologies capable of accounting for both measurable and latent effects of modifications.

Impact on Compliance Responsibilities

Another significant issue arises from the shifting of compliance responsibilities when modifications are made. Under regulatory frameworks such as the EU AI Act, certain modifications may transfer provider obligations from the original developer to the entity performing the modification. This transfer may include requirements for documentation, risk mitigation, transparency, and reporting. Determining when and to what extent these obligations apply requires careful assessment of the nature and scope of the modification.

A related challenge is the interpretation of obligations within complex value chains. AI systems are often integrated into broader applications or services that involve multiple actors, each contributing modifications or configurations. Identifying which actor assumes regulatory responsibility at each stage can be difficult, particularly when modifications are incremental, distributed, or collaborative. This complexity emphasizes the importance of clear contractual arrangements, governance structures, and documentation practices to delineate accountability.

Documentation and Traceability Challenges

Documentation is a central requirement for demonstrating compliance with AI regulations, particularly when modifications are involved. Maintaining accurate, complete, and traceable records of all changes to a system is essential to satisfy regulatory expectations. However, practical challenges arise in large or distributed development environments. Modifications may occur in parallel across teams, iterations, or automated pipelines, making comprehensive documentation resource-intensive.

Traceability is particularly important when modifications affect system outputs in ways that have regulatory or societal significance. Without proper documentation, organizations may struggle to demonstrate due diligence, respond to audits, or investigate incidents. Furthermore, incomplete traceability undermines internal governance, increases operational risk, and can reduce stakeholder confidence in the integrity of the AI system.

Technical Measurement Limitations

Assessing the impact of modifications often requires quantitative measurement of computational, statistical, or performance-related factors. For example, determining whether a modification significantly changes a model's behavior may involve evaluating changes in output distributions, accuracy metrics, or computational resources used in retraining. However, these measurements can be challenging for several reasons.

First, many AI systems are opaque, with complex internal representations that are not easily interpretable. Minor parameter adjustments may lead to disproportionate or nonlinear effects, making it difficult to quantify the modification's true impact. Second, access to baseline data or original model specifications may be limited, particularly when relying on third-party models or collaborative ecosystems. Third, traditional quantitative metrics may not capture qualitative aspects of risk, such as ethical considerations, societal impact, or potential misuse. These limitations highlight the need for integrated approaches that combine technical evaluation with broader risk and governance assessments.

Integration and Systemic Interactions

Modifications to AI systems do not occur in isolation; they often interact with other systems, processes, or operational environments. This creates the challenge of systemic interactions, where changes in one component may propagate and affect the performance, risk profile, or regulatory compliance of the entire ecosystem.

For example, modifying an AI model that interfaces with decision-support systems, automated workflows, or user-facing applications may have cascading effects. Even when a modification is intended to improve a specific function, it may inadvertently affect other outputs, introduce biases, or create vulnerabilities.

Understanding these interactions requires a holistic perspective, encompassing both technical dependencies and organizational processes. Failure to account for systemic effects can result in underestimated risks, misaligned responsibilities, and regulatory non-compliance.

Vendor Transparency and Third-Party Dependencies

In many ecosystems, organizations modify AI systems developed by third-party providers. Limited transparency regarding the original model's development, training data, and previous modifications creates significant challenges for downstream modifiers. Without access to sufficient technical documentation or operational histories, assessing the impact of modifications, evaluating risks, or meeting compliance obligations becomes difficult.

This issue is compounded in distributed or multi-vendor environments, where AI components may be sourced from multiple providers. Ensuring accountability requires proactive engagement with upstream developers, clear contractual obligations regarding documentation and transparency, and mechanisms for sharing relevant information across the value chain. Lack of clarity in these areas can lead to operational and legal uncertainties, potentially undermining the effectiveness of AI governance frameworks.

Timing and Regulatory Uncertainty

The regulatory context surrounding AI modifications is evolving. Deadlines for compliance, publication of guidelines, and establishment of standards may vary across jurisdictions or phases of implementation. Organizations modifying AI systems during periods of legal uncertainty face the challenge of aligning operational practices

with provisional guidance, without full clarity on future obligations.

Timing considerations also affect legacy systems and previously deployed AI models. Determining whether modifications to older systems trigger new obligations, or whether pre-existing models are "grandfathered" under previous standards, introduces additional complexity. Organizations must balance proactive compliance with flexibility to adapt to evolving regulatory interpretations, requiring governance frameworks that are both structured and adaptable.

Governance and Organizational Challenges

Implementing a robust governance framework to manage AI modifications is a critical, yet challenging, requirement. Effective governance involves defining roles and responsibilities, establishing review processes, documenting modifications, conducting risk assessments, and monitoring compliance. However, organizations may face barriers including resource constraints, insufficient expertise, or distributed operational models.

Coordination between technical teams, legal and compliance departments, and management is necessary to ensure that modifications are appropriately assessed and documented. Without integrated governance, organizations risk inconsistent implementation, overlooked risks, and non-compliance. Moreover, effective governance requires a culture of accountability, clear communication, and continuous adaptation to regulatory updates and technological developments.

Balancing Innovation and Compliance

A persistent conceptual challenge is balancing innovation with compliance obligations. Modifications to AI systems are often driven by operational needs, performance improvements, or new use cases. At the same time, these changes may introduce regulatory obligations that require additional documentation, risk mitigation, or oversight.

Striking an appropriate balance requires structured decision-making frameworks that assess both the technical and regulatory implications of modifications. Organizations must determine which modifications are strategically valuable, evaluate associated risks, and implement appropriate governance measures. This balance is particularly critical in rapidly evolving technological environments, where delays or overly cautious approaches can hinder innovation, while insufficient attention to compliance can result in legal, reputational, and operational risks.

Ethical and Societal Considerations

Beyond technical and regulatory dimensions, modifications to AI systems raise ethical and societal issues. Changes in AI behavior can affect fairness, transparency, accountability, and user trust. Even when modifications do not meet the regulatory threshold for substantial impact, they may introduce unintended biases or influence decision-making in ways that have societal significance.

Addressing these considerations requires organizations to integrate ethical evaluation into the modification process, including assessing potential harms, unintended consequences, and mitigation strategies. Ethical governance complements regulatory compliance and supports sustainable, responsible AI development, reinforcing

stakeholder confidence and societal acceptance.

Operational Complexity and Resource Implications

Managing modifications under regulatory frameworks introduces operational complexity and resource implications. Evaluating whether a modification is substantial, conducting risk assessments, maintaining documentation, and monitoring compliance all require dedicated personnel, technical expertise, and governance infrastructure.

Smaller organizations or teams may face particular challenges in meeting these requirements, while larger ecosystems must ensure consistent practices across multiple teams, units, or locations. Resource allocation decisions must balance operational efficiency, regulatory compliance, and innovation goals. Failure to adequately allocate resources can result in gaps in governance, increased risk exposure, and potential non-compliance.

Summary of Conceptual Challenges

In summary, modifications to AI systems under regulatory frameworks such as the EU AI Act present a complex set of issues:

- **Conceptual Ambiguity:** Difficulty in defining substantial modifications.
- **Risk Assessment:** Challenges in quantifying direct and indirect impacts.
- **Compliance Responsibility:** Shifting obligations along value chains.
- **Documentation and Traceability:** Resource-intensive record-keeping requirements.

- **Technical Measurement:** Limitations in assessing system behavior and outputs.
- **Systemic Interactions:** Cascading effects across integrated systems.
- **Vendor Transparency:** Dependence on upstream providers for information.
- **Regulatory Uncertainty:** Evolving guidance and standards.
- **Governance:** Coordination challenges across organizational functions.
- **Innovation vs. Compliance:** Balancing operational improvements with regulatory obligations.
- **Ethical Considerations:** Assessing societal impacts of modifications.
- **Operational Complexity:** Managing resources, expertise, and infrastructure effectively.

These issues collectively highlight that modifying AI systems is not merely a technical task but an integrated governance, risk management, and compliance challenge. Addressing these issues requires a combination of structured frameworks, cross-functional collaboration, and ongoing monitoring of both technological and regulatory developments.





When are you
qualified as a
provider for AI
modifications
under the EU
AI Act



The European Union's regulatory framework for artificial intelligence establishes responsibilities for entities that develop, deploy, or modify AI systems and models. Among these responsibilities, the concept of a "provider" is central, as it delineates who is accountable for compliance with obligations such as risk management, documentation, transparency, and monitoring. Determining when an organization assumes the role of a provider becomes particularly complex in the context of modifications to existing AI systems or general-purpose AI (GPAI) models. This section explores, in conceptual and analytical terms, the types of modifications that qualify an entity as a provider, the underlying criteria for assessment, and the broader implications for governance, risk management, and compliance within AI ecosystems.

Defining the Provider Role in the Context of Modifications

A provider is generally understood as an entity responsible for ensuring that an AI system meets regulatory, operational, and safety requirements. When an organization develops an AI system from the ground up, the provider role is typically clear: the developer assumes full responsibility for compliance. However, the scenario becomes more nuanced when a third party modifies an existing system or model. In such cases, the entity performing the modification may assume provider responsibilities if the changes are deemed substantial, thereby altering the AI system's behavior, generality, or risk profile.

Conceptually, the provider role in modification scenarios serves to ensure accountability for changes that could materially affect the system's outputs, potential for misuse, or societal impact. It functions as a safeguard that extends

compliance responsibilities to actors who actively shape the system beyond routine operation or minor adjustments. By framing the provider role around the notion of substantial modification, regulatory frameworks create a mechanism to allocate responsibility along complex value chains.

Conceptual Criteria for Determining Substantial Modification

Determining whether a modification triggers provider obligations requires assessing the modification's effect on three principal dimensions: **generality**, **capabilities**, and **systemic risk**. These dimensions provide a conceptual framework for evaluating whether an entity should be classified as a provider.

1. **Generality:** Changes that alter the breadth of tasks or applications a system can perform fall within the generality dimension. For instance, modifications that extend a model's functionality beyond its original scope, enabling new types of outputs or decision-making capabilities, may indicate substantial modification.
2. **Capabilities:** Modifications that materially change a system's performance, predictive accuracy, or operational characteristics are considered substantial. This includes enhancements, retraining, or structural adjustments that improve or shift the system's abilities in a way that may influence outcomes or user interactions.
3. **Systemic Risk:** Systemic risk refers to the potential for modifications to affect the broader ecosystem, including unintended consequences, cascading effects, or emergent behaviors that could influence

multiple stakeholders. Modifications that increase the likelihood of harm, bias, or operational failure elevate the risk profile and may trigger provider obligations.

The conceptual framework highlights that provider responsibilities are not solely determined by the act of modification itself but by the implications of the change. Minor or routine adjustments that do not alter these dimensions typically do not transfer provider obligations, whereas modifications with meaningful impact on any of these dimensions are more likely to do so.

Categories of Modifications

Modifications to AI systems or models can be grouped into categories to facilitate conceptual analysis and risk assessment. Each category has distinct characteristics with respect to potential provider obligations:

1. **No Change:** Using a system or model as-is, without modification, does not confer provider status. Entities in this category act as users or integrators but remain outside the provider role, unless other responsibilities are imposed through the deployment or operational context.
2. **Parameter or Hyperparameter Adjustment:** Minor tuning of operational parameters, such as thresholds, learning rates, or output scaling, typically does not qualify as substantial modification. While these adjustments may optimize performance, they do not fundamentally change the model's generality, capabilities, or systemic risk profile.
3. **Integration and Application-Level Changes:** Incorporating an AI system into a broader application or service may alter its operational context. When modifications are limited to interfacing, presentation, or orchestration, without retraining or structural change, the provider role usually remains with the original developer. However, integration that involves customization affecting performance or outputs may trigger provider obligations.
4. **Fine-Tuning:** Training a model on new datasets to adjust behavior for specific contexts is a prime example of a modification that may confer provider responsibilities. Fine-tuning can materially alter performance, influence outputs, and shift the risk profile, particularly when the dataset significantly differs from the original training set.
5. **Knowledge or Model Distillation:** Creating a derivative model based on the outputs of a larger or original model, often to reduce computational requirements or adapt for specific tasks, can constitute a substantial modification. This process changes the architecture or learned representations in a way that may affect generality, capabilities, or risk.
6. **Architectural Changes:** Structural modifications to the model, such as altering layers, components, or core algorithms, are typically substantial. Architectural changes can significantly impact the system's capabilities, introduce new behaviors, or affect systemic risk, thereby conferring provider obligations on the modifying entity.

7. **Behavioral Overrides or Parameter Manipulation:** Adjustments intended to circumvent built-in constraints or safety measures can materially affect outputs and risk profiles. These modifications are generally considered substantial because they may introduce new risks and responsibilities for the modifier.
8. **Minor Functional Enhancements:** Adding non-intrusive functionalities, such as additional output formatting or lightweight retrieval mechanisms, usually does not trigger provider obligations. The key distinction is whether these changes affect the underlying model's performance, scope, or systemic risk.

By classifying modifications into these categories, organizations can develop analytical frameworks for assessing when the provider role is assumed. Importantly, these categories are conceptual and should be applied alongside impact assessments and risk evaluation to determine regulatory responsibilities.

Thresholds and Indicators for Substantial Modification

Conceptual frameworks often include thresholds or indicators to guide the determination of provider obligations. These thresholds may be based on quantitative measures, qualitative assessment, or a combination of both. Key considerations include:

- **Computational Effort:** The resources required to implement the modification, such as retraining or fine-tuning, can serve as an indicator of substantial modification. While computational thresholds are indicative rather than definitive, higher resource use may correlate

with meaningful changes in model behavior or capability.

- **Change in Output Behavior:** Modifications that produce significantly different outputs compared to the original system are more likely to be substantial. Quantitative measures, such as changes in accuracy, prediction distributions, or error rates, can support this assessment.
- **Impact on Risk Profile:** Modifications that increase operational, ethical, or systemic risks, including potential for misuse, bias, or failure, serve as strong indicators of substantial modification. Qualitative analysis and scenario evaluation are often necessary to capture these effects.
- **Scope Expansion:** Extending the range of tasks or applications for which the system can be used is a conceptual threshold. Modifications enabling new use cases beyond the original design may qualify as substantial.

These thresholds function as practical tools within a conceptual framework, assisting organizations in evaluating modifications against criteria for provider obligations. They highlight that both technical and contextual factors must be considered in determining regulatory responsibility.

Assessment Methodology for Modifications

A structured methodology is essential for evaluating whether modifications confer provider status. The following steps are broadly applicable across ecosystems:

1. **Identification of Modifications:** Catalog all changes made to the AI system or model, including retraining, parameter adjustments,

integration steps, and structural alterations. Accurate identification provides the foundation for subsequent analysis.

2. **Evaluation of Impact on Generality, Capabilities, and Risk:** Assess whether each modification affects the model's scope, performance, or systemic risk. Both qualitative and quantitative methods should be employed to capture the full range of effects.
3. **Comparison Against Indicators and Thresholds:** Evaluate modifications against conceptual thresholds, such as computational effort, output deviation, or potential for harm. This helps determine the relative significance of each change.
4. **Documentation and Justification:** Maintain detailed records of modifications, the assessment process, and the rationale for conclusions regarding provider obligations. Documentation supports transparency, governance, and regulatory compliance.
5. **Decision on Provider Status:** Based on the assessment, determine whether the entity performing the modifications assumes provider responsibilities. This decision informs risk management, governance, and operational practices.
6. **Continuous Monitoring and Reassessment:** Modifications may have evolving effects over time, particularly in adaptive or dynamic AI systems. Ongoing monitoring ensures that provider obligations remain aligned with system behavior and risk profile.

Conceptual Implications for AI Governance

Recognizing the types of modifications that confer provider responsibilities has broader implications for governance and organizational strategy.

Key considerations include:

- **Role Clarity:** Clear delineation of provider responsibilities along the value chain reduces ambiguity, supports accountability, and ensures regulatory compliance.
- **Risk Management:** Assessing and documenting modifications facilitates proactive identification and mitigation of risks associated with AI system outputs, behavior, or integration.
- **Resource Allocation:** Determining when provider status is assumed allows organizations to allocate governance, legal, and technical resources efficiently, ensuring that substantial modifications are managed with appropriate oversight.
- **Transparency and Traceability:** Conceptual frameworks for modifications support the creation of transparent records, enhancing trust with regulators, stakeholders, and end-users.
- **Strategic Decision-Making:** Understanding which modifications trigger provider obligations informs organizational strategy, including decisions on whether to modify systems internally, rely on third-party providers, or adopt alternative deployment strategies.

Challenges in Multi-Actor Ecosystems

AI systems are often developed, deployed, and modified in multi-actor ecosystems involving upstream providers, downstream integrators, and end-users. Identifying which actor assumes provider responsibilities can be challenging when modifications are incremental, collaborative, or distributed.

Conceptually, the following factors must be considered:

- **Nature of Contribution:** Whether an actor's modifications materially influence generality, capabilities, or risk profile. Minor operational adjustments may leave provider obligations with the upstream developer, whereas substantial alterations shift responsibilities.
- **Integration Context:** Modifications that occur within broader applications or systems may have cascading effects. The provider role may depend on the scope of these effects and the degree of control the modifying entity has over outputs and behavior.
- **Coordination and Contractual Clarity:** Multi-actor ecosystems require agreements that define roles, responsibilities, and documentation obligations. Conceptual clarity regarding provider status ensures accountability and reduces disputes.
- **Transparency and Access to Information:** Modifiers need sufficient information from upstream developers to assess impacts. Limited transparency can complicate the determination of

provider responsibilities, highlighting the importance of shared governance and information exchange.

Summary and Conceptual Insights

Determining which modifications qualify an entity as a provider under the EU AI Act requires a structured, analytical, and conceptually grounded approach. Key insights include:

- The provider role is tied to **substantial modifications** that meaningfully affect generality, capabilities, or systemic risk.
- Modifications vary from **minor parameter adjustments** to **fine-tuning, structural changes, or behavioral overrides**, with only the latter generally conferring provider obligations.
- Conceptual thresholds, including computational effort, output behavior, and risk profile, provide guidance but require contextual interpretation.
- Structured assessment methodologies, including identification, evaluation, documentation, and monitoring, support consistent determination of provider status.
- Multi-actor ecosystems introduce complexity in allocating provider responsibilities, emphasizing the need for governance framework.
- Understanding the implications of modifications informs governance, resource allocation, risk management, and strategic decision-making across organizations and ecosystems.



General- Purpose AI in Action: A Practitioner's Challenge



General-purpose artificial intelligence (GPAI) has emerged as a transformative class of AI systems capable of performing a wide range of tasks across multiple domains. Unlike task-specific models, which are designed for narrowly defined purposes, GPAI systems exhibit adaptability and flexibility, allowing them to address diverse challenges through a single architecture. These systems operate by processing inputs in natural or structured formats, generating outputs that support decision-making, problem-solving, or content creation across contexts. The practical deployment of GPAI systems has rapidly expanded, with organizations integrating them into operational workflows, product offerings, and research activities. This section explores conceptual examples of GPAI in practice and examines the open challenges faced by practitioners in modifying, managing, and governing these systems within complex operational ecosystems.

Conceptual Practitioner Examples

While the following examples are generalized and abstracted from specific technologies, they illustrate the breadth of GPAI applications and the types of considerations practitioners encounter in real-world use.

Integrated Knowledge and Decision Support Platforms

Organizations often leverage GPAI systems to aggregate, process, and synthesize information from multiple sources to support decision-making. In such platforms, the GPAI system acts as a central analytical engine, enabling users to query large datasets, generate summaries, or simulate potential outcomes. Practitioners deploying these systems may focus on integrating organizational knowledge bases, defining

data access policies, and configuring interaction workflows. Modifications to the GPAI system in this context often involve prompt design, embedding specialized knowledge, or creating interfaces that translate raw outputs into actionable insights. While these adjustments may be operationally minor, they can significantly affect the system's perceived reliability and utility.

Agentic Systems for Workflow Automation

Another conceptual example involves GPAI-driven agentic systems that execute tasks autonomously or semi-autonomously within defined workflows. In these configurations, the GPAI system interacts with digital environments, other software tools, or user inputs to perform iterative tasks, such as research synthesis, automated reporting, or process orchestration. Practitioners modifying GPAI systems for such workflows focus on task sequencing, constraint definition, and integration with auxiliary tools. Even when the core model remains unaltered, modifications to the orchestration logic can influence system behavior and outputs. Evaluating these effects conceptually requires understanding both the model's capabilities and the broader operational context in which it functions.

Customization for Context-Specific Applications

Organizations frequently adapt GPAI systems to address context-specific requirements without fundamentally altering the underlying architecture. Customization may include fine-tuning prompts, integrating domain-specific knowledge sources, or adjusting the system's interface to reflect user needs. Such modifications are intended to optimize relevance, interpretability, and

user experience. Practitioners implementing these adaptations must carefully monitor whether changes materially affect outputs or introduce biases, as these factors influence operational reliability and governance responsibilities. Conceptually, these examples highlight the spectrum of modification intensity, from lightweight operational adjustments to more substantial fine-tuning.

Analytical Observations on GPAI Modifications

The practitioner examples underscore several analytical considerations relevant to GPAI deployment and modification. These considerations focus on system behavior, governance, and risk management across operational ecosystems.

Modifications and System Behavior

Even seemingly minor modifications to GPAI systems can alter outputs in unanticipated ways. For instance, adjustments to input processing pipelines, integration of auxiliary tools, or prompt engineering may result in changes to consistency, specificity, or bias in generated outputs. Practitioners must recognize that the behavior of GPAI systems emerges from complex interactions between the model, data inputs, and operational environment. Consequently, analytical assessment of modifications requires both qualitative evaluation—such as understanding potential societal impacts—and quantitative evaluation, including monitoring changes in output distributions, error rates, or performance metrics.

Assessing Operational Impact

Practitioners must evaluate how modifications affect system outputs in operational workflows. In integrated

knowledge platforms, even minor contextual adjustments can influence decision-support outputs, potentially affecting organizational decisions. Similarly, in agentic workflows, modifications to task orchestration logic may change the timing, prioritization, or sequencing of outputs, with downstream effects on operational efficiency and quality. Analytical frameworks for assessing operational impact consider both direct effects on outputs and indirect effects on users, systems, or stakeholders that interact with the GPAI system.

Maintaining Adaptability

GPAI systems are inherently flexible, capable of supporting a wide array of tasks.

Practitioners modifying these systems must balance optimization for specific use cases with the preservation of general-purpose capabilities. Over-specialization may limit the system's adaptability, reducing its value across broader applications. Analytical evaluation therefore involves assessing trade-offs between contextual optimization and general-purpose functionality. Modifications that improve performance in one domain may unintentionally degrade capabilities in others, necessitating ongoing monitoring and evaluation.

Conceptual Challenges in GPAI Deployment

The practical deployment of GPAI systems introduces a set of conceptual and analytical challenges that extend beyond technical optimization. These challenges span operational, organizational, and governance dimensions and have implications for risk management, accountability, and compliance.

Ethical and Governance Considerations in AI Deployment

Beyond operational challenges, practitioners face critical ethical and governance

considerations that directly impact deployment reliability. Recent research on AI's role in strategic decision-making highlights three priority areas:

1. **Algorithmic Bias & Fairness:** The risk of amplifying historical biases requires fairness-aware algorithms and regular bias impact assessments.
2. **Explainable AI (XAI):** The "black box problem" demands transparency mechanisms to build stakeholder trust, particularly for high-stakes decision support.
3. **Data Security:** Encryption, access controls, and ML-based anomaly detection are essential to protect sensitive data processed by modified GPAI systems.

These governance dimensions are particularly critical when modifications occur in multi-actor ecosystems where downstream modifiers assume provider responsibilities under the EU AI Act.

Complexity of Risk Assessment

GPAI systems exhibit complex behavior that may be sensitive to small modifications. Assessing risk in these systems involves understanding potential operational failures, biases, or unintended consequences. In practice, even low-level modifications, such as integrating external knowledge sources or adjusting prompts, can create emergent risks that are difficult to quantify. Practitioners face the challenge of developing risk assessment methodologies that capture both measurable and latent effects, including ethical considerations, societal impacts, and systemic interactions with other processes or systems.

Accountability and Provider Responsibilities

In multi-actor ecosystems, determining accountability for GPAI modifications is conceptually challenging. Organizations may act as downstream integrators, system modifiers, or operational users, with overlapping responsibilities. Modifications that materially affect system behavior can trigger provider responsibilities, encompassing obligations such as documentation, transparency, and monitoring. Analytical frameworks are required to evaluate the magnitude of modifications, the nature of their effects, and the corresponding allocation of responsibility across actors. Clear delineation of roles is essential for governance, compliance, and risk mitigation.

Documentation and Traceability

Effective governance of GPAI modifications relies on maintaining comprehensive documentation and traceability. Practitioners must record the nature, purpose, and impact of modifications, including changes to prompts, input pipelines, task orchestration logic, or retraining processes. Conceptually, traceability supports accountability, risk assessment, and monitoring, enabling organizations to respond to incidents or evaluate system performance over time. The challenge lies in implementing documentation processes that are both rigorous and operationally feasible, particularly in distributed or collaborative environments.

Systemic Interactions and Emergent Behavior

GPAI systems often operate within interconnected ecosystems. Modifications to one component may propagate through

dependent systems, producing emergent behaviors that were not present in the original configuration. Practitioners must adopt a systemic perspective, evaluating not only the direct effects of modifications but also potential cascading impacts.

Conceptually, this requires understanding both the technical dependencies and the operational contexts in which the GPAI system interacts with other processes.

Multi-Layered Risk Assessment Framework

Research on AI governance frameworks identifies that comprehensive risk assessment must encompass not only technical dimensions (model drift, accuracy changes) but also ethical and organizational factors.

Comprehensive risk assessment must extend beyond technical metrics to encompass ethical and organizational dimensions.

Effective governance requires a 3-layer risk framework:

- a. fairness-aware algorithm design with regular bias impact assessments
- b. organizational readiness across technical and cultural dimensions and
- c. robust data protection through encryption, access controls, and ML-based anomaly detection.

This multi-layered approach addresses both measurable risks and latent ethical implications.

Balancing Customization and Generality

Modifications often aim to enhance the system's relevance to specific tasks or contexts. However, extensive customization can reduce the general-purpose capabilities that define GPAI systems. Practitioners must balance the benefits of task-specific optimization with the need to preserve broader adaptability. Analytical assessment

involves evaluating the impact of modifications on both specialized outputs and general-purpose functionality, considering potential trade-offs and long-term operational implications.

Conceptual Approaches to Managing GPAI Modifications

To address the challenges outlined, practitioners adopt structured approaches to managing modifications, emphasizing risk assessment, governance, and operational oversight.

Risk-Based Modification Assessment

A central component of modification management is risk-based assessment. Practitioners evaluate modifications along dimensions of potential operational, ethical, and systemic impact. Conceptually, this approach involves identifying modifications, analyzing their effect on outputs and stakeholder interactions, and determining whether the changes introduce new risks or amplify existing ones. Risk assessment frameworks support prioritization of modifications, guiding governance resources toward changes with the greatest impact.

Governance and Accountability Frameworks

Effective management of GPAI modifications requires clear governance structures. Conceptual frameworks define roles, responsibilities, and decision-making processes for system modifications, enabling organizations to allocate provider responsibilities appropriately. Governance frameworks also facilitate coordination across technical, legal, and operational teams, ensuring that modifications are evaluated consistently, documented comprehensively, and monitored for ongoing impact.

Documentation and Change Tracking

Practitioners implement structured documentation and change tracking to support transparency and accountability. Conceptual approaches include maintaining modification logs, technical summaries, and impact assessments. These records enable retrospective evaluation, support internal audits, and facilitate compliance with external oversight or reporting requirements. The emphasis is on traceability and completeness rather than procedural rigidity, ensuring that modifications can be assessed objectively and systematically.

Monitoring and Continuous Evaluation

Given the dynamic nature of GPAI systems, continuous monitoring of modifications is essential. Conceptually, this involves evaluating outputs, assessing system performance, and identifying emergent behaviors over time. Monitoring frameworks may include automated performance metrics, error analysis, and feedback loops for iterative improvement. Continuous evaluation allows organizations to detect unintended consequences of modifications and adapt governance practices accordingly.

Open Challenges and Areas for Further Exploration

Despite structured approaches, practitioners face open challenges in deploying and managing GPAI systems. These challenges are conceptual and broadly applicable across ecosystems.

Quantifying Substantial Modifications

Determining which modifications materially affect system behavior remains an open challenge. While thresholds based on computational effort, output deviation, or risk indicators are conceptually useful, they often require contextual interpretation. Practitioners must balance quantitative measures with qualitative judgment, considering the operational and societal implications of modifications.

Balancing Innovation and Governance

Organizations face tension between innovation and structured oversight. GPAI modifications may drive performance improvements or new applications but also introduce regulatory, operational, or ethical risks. Conceptually, practitioners must adopt frameworks that support iterative innovation while ensuring accountability, transparency, and risk management.

Multi-Actor Responsibility Allocation

In ecosystems with multiple actors modifying and integrating GPAI systems, assigning responsibilities is complex. Conceptual challenges include identifying when downstream actors assume provider responsibilities, managing contractual obligations, and coordinating governance practices across diverse teams or organizations.

Emergent and Systemic Effects

Modifications may produce emergent behaviors that are difficult to predict. Practitioners must consider interactions with other systems, user inputs, and operational

contexts, evaluating both direct and indirect consequences. Systemic effects pose analytical challenges, as small changes may propagate in nonlinear ways, affecting outputs, stakeholders, or ecosystem stability.

Documentation and Transparency in Practice

Maintaining comprehensive, actionable documentation remains operationally demanding. Conceptually, organizations must balance completeness with practicality, ensuring that records support governance, risk assessment, and accountability without impeding operational efficiency.

Ethical and Societal Considerations

Even when modifications do not trigger formal obligations, they may affect fairness, equity, transparency, or societal outcomes. Practitioners must incorporate ethical evaluation into modification workflows, considering potential unintended consequences and mitigation strategies. Conceptual frameworks that integrate ethics, governance, and operational management are essential for responsible deployment.

Synthesis and Conceptual Insights

The practitioner examples and open challenges described illustrate several key insights about GPAI in action:

1. **Modification intensity varies widely:** From lightweight prompt adjustments to fine-tuning or architectural changes, modifications differ in their operational and regulatory significance.
2. **Impact on outputs and behavior is central:** Modifications are conceptually significant when they

materially affect system behavior, performance, or risk profile.

3. **Governance structures are critical:** Effective deployment requires clear accountability, structured documentation, and risk assessment processes.
4. **Ecosystem context matters:** Modifications occur within complex multi-actor environments, necessitating consideration of dependencies, systemic interactions, and coordination.
5. **Ethical and societal considerations are integral:** Conceptual frameworks must integrate operational, technical, and ethical perspectives to support responsible GPAI deployment.
6. **Continuous monitoring is essential:** Adaptive and dynamic systems require ongoing evaluation to detect emergent behaviors, unintended consequences, and evolving risk profiles.

Collectively, these insights emphasize that GPAI deployment is not purely a technical exercise. It involves integrated consideration of operational context, risk, governance, and ethical implications. Practitioner approaches that recognize the interplay between modification intensity, systemic effects, and accountability provide a foundation for responsible, transparent, and effective utilization of general-purpose AI systems.

Conceptual Takeaways for Future Practice

From a conceptual and analytical perspective, the study of GPAI in action highlights several considerations for organizations and stakeholders:

- **Structured Assessment:** Establishing frameworks for evaluating modifications along dimensions of impact, risk, and adaptability ensures systematic decision-making.
- **Documentation and Traceability:** Maintaining comprehensive records supports transparency, accountability, and operational oversight.
- **Adaptive Governance:** Governance structures must be flexible to accommodate the evolving behavior of GPAI systems and the complexity of modifications.
- **Risk Management:** Analytical approaches to assessing potential operational, ethical, and systemic risks are essential to informed decision-making.
- **Ethical Integration:** Ethical evaluation should be embedded within modification workflows, addressing potential societal and stakeholder impacts.
- **Multi-Actor Coordination:** Effective deployment in collaborative or distributed ecosystems requires clear delineation of responsibilities and shared governance practices.

These conceptual takeaways provide guidance for organizations seeking to implement GPAI systems responsibly, balancing innovation, operational effectiveness, and accountability.



Case Study: Third-Party LLM integration for internal- decision support

Consider a hypothetical enterprise software provider that integrates a third-party general-purpose language model into an internal decision-support tool. Initially, the model is used “as-is,” with prompt engineering and retrieval-augmented generation to reference internal documentation. At this stage, the organization documents the integration but does not assume provider obligations.

Over time, performance concerns lead the organization to fine-tune the model using a large proprietary dataset derived from internal operational records. The fine-tuning process consumes significant computational resources and produces measurable changes in output style, confidence, and decision recommendations. A structured assessment is initiated.

First, technical teams document the modification, including datasets used, training methods, and observed behavioural changes. Second, a risk assessment evaluates whether outputs now influence regulated decisions and whether bias or reliability risks have increased. Third, governance teams assess whether the modification alters the model’s intended purpose and risk profile relative to the original system.

Based on this assessment, the organization concludes that the fine-tuning constitutes a substantial modification. As a result, it assumes provider responsibilities under the EU AI Act, updating documentation, implementing enhanced monitoring, and revising internal accountability structures.

This hypothetical illustrates how technical measurement, risk assessment, documentation, and governance interact in practice—and how provider status can shift as modifications move beyond operational customization into substantive system change.



Andrea Frosinini,
Business Development
Manager, XDC
Foundation



Going
Forward



As artificial intelligence continues to evolve and integrate into organizational processes, it is increasingly critical for organizations to establish robust frameworks for governance, compliance, and risk management. Regulatory frameworks, such as those focused on general-purpose AI, emphasize the responsibilities of entities that develop, deploy, or modify AI systems. Organizations operating within such frameworks must consider not only technical and operational requirements but also strategic, ethical, and systemic implications. This section provides general recommendations for organizations seeking to manage AI effectively and outlines a set of recommended actions and resources to support preparation for regulatory compliance. The discussion is conceptual, analytical, and broadly applicable to organizations across diverse ecosystems.

General Recommendations for Organizations

Organizations integrating AI systems face a complex landscape of opportunities, risks, and responsibilities. General recommendations focus on establishing foundational capabilities, fostering a culture of accountability, and aligning operational practices with organizational and societal objectives.

Establish a Structured AI Governance Framework

A governance framework provides the organizational scaffolding for managing AI systems, modifications, and associated risks. Conceptually, an effective governance framework should:

- **Define Roles and Responsibilities:** Clearly delineate the roles of developers, operators, integrators, and decision-makers within the AI lifecycle. Role clarity ensures

accountability for design, modification, deployment, and monitoring activities.

- **Integrate Risk Assessment**
Mechanisms: Incorporate systematic procedures to identify, evaluate, and mitigate operational, ethical, and systemic risks associated with AI. Risk assessments should account for both the technical properties of AI systems and their broader operational context.
- **Support Decision-Making:** Provide structures and processes that inform executive decisions on AI adoption, modification, and deployment. Decision-making should balance innovation, operational efficiency, and risk mitigation.
- **Facilitate Continuous Improvement:** Embed processes for ongoing evaluation and refinement of governance practices, allowing organizations to adapt to evolving AI capabilities, operational requirements, and regulatory expectations.

A structured AI governance framework serves as the foundation for aligning technical practices with organizational strategy and regulatory compliance. It ensures that AI initiatives are not only operationally effective but also systematically managed.

Promote Risk-Aware Culture and Organizational Awareness

Operationalizing AI responsibly requires cultivating a risk-aware culture within the organization. Key considerations include:

- **Education and Training:** Provide training programs for technical, operational, and managerial staff to

increase awareness of AI risks, system behavior, and governance responsibilities.

- **Cross-Functional Collaboration:** Foster collaboration between technical teams, risk management, legal counsel, and compliance departments. Multi-disciplinary engagement ensures comprehensive assessment of modifications and operational impacts.
- **Scenario-Based Analysis:** Encourage staff to engage in scenario planning exercises to anticipate potential failures, emergent behaviors, or ethical concerns arising from AI modifications.
- **Transparent Communication:** Promote transparent internal communication regarding AI modifications, decisions, and governance practices, enabling informed operational decisions and accountability.

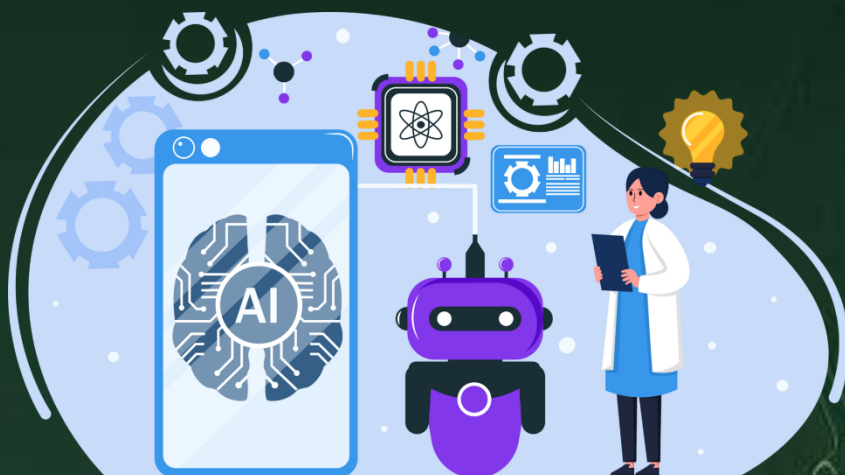
A risk-aware culture supports proactive identification of challenges, reduces operational blind spots, and strengthens the organization's capacity to respond to evolving AI capabilities and regulatory expectations.

Organisations preparing for EU AI Act compliance should adopt adaptive governance frameworks that balance innovation with structured oversight. Academic research on AI governance emphasizes that forward-looking practices require: (1) Clear delineation of stakeholder roles (developers, operators, modifiers) with transparent accountability structures; (2) Comprehensive documentation linked to risk assessments and modification histories (supporting audit readiness); and (3) Regular third-party audits with ethical safeguards embedded throughout AI lifecycles. These governance principles are particularly critical in multi-vendor ecosystems where responsibility boundaries may be ambiguous

Implement Robust Documentation and Change Management Practices

Documentation and change management are essential for accountability, traceability, and operational control. Conceptually, organizations should:

- **Document Modifications Systematically:** Maintain detailed records of changes to AI systems or models, including the rationale, technical specifications, data sources, and expected outcomes.
- **Track Operational Impacts:** Record the effects of modifications on outputs, performance metrics, and user interactions. This enables retrospective evaluation and continuous improvement.
- **Standardize Change Management Processes:** Establish formal procedures for approving, implementing, and reviewing modifications, ensuring consistency and governance oversight.



- **Enable Audit and Compliance Readiness:** Structured documentation supports internal audits, regulatory reporting, and demonstration of due diligence in modification practices.

Comprehensive documentation and change management practices provide both operational and regulatory value, enabling organizations to understand, control, and communicate the effects of AI modifications.

Conduct Regular Risk and Impact Assessments

AI systems operate in complex and dynamic environments. Continuous assessment of risks and impacts is essential for maintaining operational reliability and ethical standards. Organizations should:

- **Perform Pre-Deployment Assessments:** Evaluate potential risks before introducing AI systems or modifications into operational workflows.
- **Monitor Post-Deployment Performance:** Assess outputs, behaviors, and systemic interactions to identify emergent risks or unintended consequences.
- **Evaluate Broader Ecosystem Impacts:** Consider potential effects on interconnected systems, organizational processes, and stakeholders.
- **Integrate Quantitative and Qualitative Analysis:** Combine statistical performance metrics with ethical and operational evaluation to achieve holistic assessments.

Regular risk and impact assessments support adaptive governance, enabling organizations to mitigate operational and ethical

challenges while sustaining system performance and integrity.

Foster Transparency and Accountability

Transparency and accountability are core principles of responsible AI management. Conceptually, organizations should:

- **Provide Clear Operational Guidelines:** Ensure that internal teams understand how AI systems are deployed, modified, and monitored.
- **Establish Reporting Mechanisms:** Create channels for reporting incidents, anomalies, or unexpected outcomes associated with AI systems.
- **Maintain Traceable Records:** Ensure that all modifications, assessments, and operational decisions are documented and accessible for review.
- **Communicate with Stakeholders:** Transparently share relevant information with stakeholders, including management, operational teams, and, where appropriate, end-users or external auditors.

Transparency and accountability practices enhance trust, facilitate compliance, and enable informed decision-making across organizational layers.

Adopt Adaptive and Forward-Looking Practices

AI systems and regulatory environments are dynamic. Organizations should cultivate adaptability through:

- **Continuous Learning:** Update governance and operational practices in response to evolving AI

capabilities, industry best practices, and emerging regulatory guidance.

- **Scenario Planning:** Anticipate future operational and regulatory challenges, including potential modifications that may alter system behavior or risk profiles.
- **Strategic Resource Allocation:** Prioritize governance, compliance, and technical resources to address high-impact areas while maintaining flexibility for innovation.
- **Engagement with External Knowledge Sources:** Monitor academic research, industry initiatives, and regulatory developments to integrate insights into organizational practices.

Forward-looking practices position organizations to respond effectively to both technological evolution and regulatory changes, minimizing operational disruption and compliance risk.

Recommended Actions for AI Act Preparation

Preparing for regulatory frameworks governing AI systems requires a systematic approach that combines assessment, documentation, governance, and stakeholder engagement. The following conceptual actions provide guidance for organizations seeking to ensure readiness and operational compliance.

2.1 Catalogue and Classify AI Systems and Use Cases

The foundation of regulatory preparation is a thorough understanding of the AI systems and use cases within the organization. Recommended steps include:

- **Inventory AI Systems:** Identify all AI systems and models in use, including third-party, in-house, and integrated solutions.
- **Classify Systems by Functionality:** Categorize systems according to operational roles, technical characteristics, and potential impact on stakeholders.
- **Assess Modification Potential:** Evaluate the degree to which systems can be adapted, modified, or extended, including potential effects on outputs and risk profiles.
- **Identify High-Risk Applications:** Conceptually assess which systems or use cases may pose higher operational, ethical, or systemic risks.

A comprehensive inventory and classification process provides the analytical foundation for risk management, governance, and compliance assessment.

2.2 Conduct Preliminary Risk and Impact Assessments

Organizations should establish structured assessment practices to evaluate the potential implications of AI systems and modifications. Key elements include:

- **Operational Risk Analysis:** Examine potential technical failures, accuracy limitations, and system reliability concerns.
- **Ethical and Societal Considerations:** Analyze risks related to fairness, transparency, privacy, and stakeholder impact.

- **Systemic and Interdependency Assessment:** Evaluate how modifications may influence interconnected systems or operational processes.
- **Scenario Testing and Simulation:** Use conceptual modeling or simulation exercises to explore potential outcomes under different conditions or modifications.

Preliminary risk assessments support informed decision-making regarding system deployment, modification, and ongoing governance.

2.3 Implement Structured Documentation Practices

Documentation supports both operational oversight and regulatory compliance. Recommended actions include:

- **Record Modification Details:** Document technical changes, rationale, datasets, parameter adjustments, and expected outcomes.
- **Capture Operational Context:** Record how AI systems interact with workflows, end-users, and other systems.
- **Maintain Version Control:** Track iterative modifications to ensure traceability and accountability.
- **Integrate Assessment Findings:** Link risk, impact, and ethical assessments to corresponding modifications and operational decisions.

Structured documentation ensures organizations can demonstrate due diligence, facilitate audits, and maintain internal and external accountability.

2.4 Develop and Maintain Governance Frameworks

A comprehensive governance framework aligns AI operations with strategic objectives, regulatory expectations, and ethical standards. Recommended actions include:

- **Define Organizational Roles:** Clarify provider, operator, and modifier responsibilities across AI lifecycles.
- **Integrate Risk and Impact Review:** Establish formal processes for evaluating modifications, assessing system outputs, and monitoring ongoing risks.
- **Incorporate Monitoring and Feedback Loops:** Enable continuous performance evaluation and detection of emergent behaviors.
- **Align Policies with Operational Practices:** Ensure that internal policies reflect technical, operational, and ethical realities of AI deployment.

Governance frameworks provide the structural foundation for consistent, accountable, and risk-informed AI operations.

2.5 Establish Monitoring and Evaluation Mechanisms

Ongoing monitoring is essential to understand the effects of modifications and maintain system integrity. Recommended actions include:

- **Define Performance Metrics:** Identify indicators of system reliability, accuracy, and effectiveness.
- **Monitor Behavioral Consistency:** Evaluate outputs for deviation from

expected behaviors or systemic objectives.

- **Detect Emergent Risks:** Implement procedures to identify unintended consequences or operational anomalies.
- **Adapt Operational Practices:** Use monitoring insights to refine system use, modification practices, and governance protocols.

Monitoring mechanisms support proactive risk management, informed decision-making, and adaptive governance.

2.6 Engage in Continuous Knowledge Development

Organizations should remain informed regarding regulatory developments, best practices, and technological advances. Recommended actions include:

- **Track Regulatory Guidance:** Monitor updates, guidelines, and interpretative materials provided by regulatory authorities or industry bodies.
- **Participate in Peer Networks:** Engage in cross-industry knowledge-sharing initiatives to learn from the experiences of other organizations.
- **Invest in Training and Skill Development:** Enhance the competencies of staff responsible for AI development, modification, governance, and compliance.
- **Leverage External Expertise:** Incorporate insights from legal, technical, and ethical advisors to complement internal capabilities.

Continuous knowledge development ensures that organizations remain aligned

with evolving expectations and emerging operational or regulatory challenges.

2.7 Prepare for Audits and External Oversight

Conceptually, organizations must anticipate potential evaluation by regulatory bodies or external auditors. Recommended actions include:

- **Maintain Accessible Documentation:** Ensure that modification logs, risk assessments, and governance records are structured and retrievable.
- **Conduct Internal Audits:** Periodically review compliance and operational practices to identify gaps or areas for improvement.
- **Simulate Regulatory Scenarios:** Use conceptual exercises to anticipate potential inquiries or evaluation requirements.
- **Establish Communication Channels:** Ensure readiness to respond to external stakeholders with clear, accurate, and timely information.

Preparation for audits strengthens operational accountability and positions organizations to demonstrate regulatory readiness.

3. Conceptual Resources for Preparation

Organizations can leverage a variety of conceptual tools, frameworks, and resources to support AI system management and regulatory preparation:

- **Compliance Checklists and Frameworks:** Structured guides that map organizational practices to regulatory expectations, helping to identify gaps and prioritize actions.
- **Risk and Impact Assessment Templates:** Conceptual frameworks for systematically evaluating potential operational, ethical, and systemic risks.
- **Governance Models and Role Matrices:** Tools that clarify responsibilities, decision-making authority, and accountability structures across the organization.
- **Documentation and Change Management Systems:** Platforms or processes that support structured record-keeping, version control, and traceability.
- **Monitoring and Evaluation Dashboards:** Analytical tools that track system performance, detect anomalies, and facilitate continuous improvement.
- **Scenario Planning Guides:** Conceptual resources for simulating potential operational or regulatory challenges and identifying mitigation strategies.
- **Knowledge-Sharing Networks:** Participation in peer groups, consortiums, or industry initiatives that facilitate exchange of insights and best practices.

Leveraging these conceptual resources enables organizations to operationalize AI governance, prepare for compliance, and manage modification-related responsibilities systematically.

4. Integrating Conceptual Insights into Organizational Practice

Organizations preparing for regulatory compliance and operational governance must integrate conceptual insights into actionable practices. Key integration points include:

- **Linking Risk Assessment to Governance:** Ensure that identification of potential risks informs operational decision-making and allocation of provider responsibilities.
- **Aligning Documentation with Monitoring:** Documentation practices should support monitoring, evaluation, and traceability, creating feedback loops for continuous improvement.
- **Embedding Adaptability into Processes:** Modify governance and operational frameworks to accommodate evolving AI capabilities, ecosystem dependencies, and regulatory expectations.
- **Fostering Organizational Awareness:** Promote cultural understanding of AI risks, modification implications, and operational responsibilities across all relevant teams.
- **Balancing Innovation and Compliance:** Establish processes that allow experimentation and iterative improvement while ensuring structured oversight and risk mitigation.

Conceptually, the integration of governance, risk assessment, documentation, and continuous monitoring forms a coherent

5. Conceptual Summary and Analytical Takeaways

The recommendations and actions outlined above provide a structured, analytical approach for organizations navigating the complexities of AI integration, modification, and regulatory preparation. Key conceptual insights include:

1. **Governance as Foundational:**
Structured frameworks for oversight, accountability, and risk management underpin all organizational AI practices.

ecosystem for responsible AI management, adaptable to various operational contexts and evolving regulatory requirements.

2. **Risk and Impact Awareness:**
Continuous evaluation of operational, ethical, and systemic risks is essential to responsible AI management.
3. **Documentation and Traceability:**
Comprehensive and accessible records enable accountability,



Governance Tools and AI Supply Chain Mapping

As AI systems increasingly rely on complex ecosystems of upstream models, downstream integrations, and third-party services, effective governance depends on visibility across the entire AI supply chain. Mapping this supply chain is particularly important for tracking modifications, allocating responsibilities, and maintaining compliance in multi-vendor environments.

AI supply chain mapping involves identifying all actors, components, and dependencies involved in the lifecycle of an AI system. This includes foundation or GPT model providers, fine-tuning vendors, data suppliers, infrastructure providers, application developers, and deployment environments. Each link in the chain may introduce modifications—intentional or incidental—that affect system behaviour or risk profile.

Governance tools can support this mapping by maintaining structured inventories of AI systems and associated components. Such tools typically record model provenance, modification history, contractual roles, and responsibility boundaries.

When modifications occur, these inventories enable organizations to trace where changes were introduced, who performed them, and whether they may trigger provider obligations under the EU AI Act.

In practice, supply chain mapping also supports risk management and procurement decisions. Organizations can assess where they depend on opaque third-party models, identify gaps in documentation or transparency, and require appropriate disclosures from vendors. As regulatory scrutiny increases, the ability to demonstrate control and oversight across the AI supply chain will become a core governance capability rather than an optional compliance exercise.



Emerging Standards and Protocols for Modification Traceability

Looking ahead, emerging technical standards and interoperability protocols may play an important role in standardising how AI modifications are recorded, communicated, and assessed across ecosystems. One area of development concerns protocols designed to structure interactions between models, tools, and contextual data, with built-in mechanisms for traceability and logging.

For example, emerging approaches such as the Model Context Protocol (MCP) point toward a future in which AI systems expose standardized interfaces for tracking how context, prompts, tools, and external resources influence model behaviour. While not a regulatory instrument, such protocols could support consistent logging of modification-relevant information, including configuration changes, contextual dependencies, and operational constraints.

If widely adopted, these protocols could enhance transparency across organizational and vendor boundaries, making it easier to demonstrate how an AI system has evolved over time. From a compliance perspective, standardized modification logs could support documentation obligations, audit readiness, and clearer allocation of responsibilities when systems are adapted downstream.

Importantly, these developments remain complementary to regulatory requirements rather than substitutes for them. However, they illustrate how technical standards may evolve to align with governance needs, reducing friction between innovation and compliance. Organizations that monitor and engage with emerging standards are likely to be better positioned to implement scalable, future-proof AI governance practices under the EU AI Act and beyond.



Andrea Frosinini,
Business Development
Manager, XDC Foundation

Conclusion

The integration and modification of advanced artificial intelligence systems represents a significant area of interest for organizations, practitioners, and stakeholders across a wide range of operational contexts. These systems, characterized by their flexibility, adaptability, and broad functional capabilities, have the potential to deliver substantial benefits in terms of efficiency, decision support, knowledge synthesis, and process automation. By enabling organizations to analyze complex information, generate insights, and optimize workflows, AI technologies can support more informed and timely decision-making, enhance operational effectiveness, and contribute to innovation. The purpose of this report has been to examine the conceptual and practical considerations surrounding the adaptation and management of such systems, highlighting their role within broader operational, organizational, and strategic contexts. Recognizing the significance of AI systems in contemporary organizational environments underscores the importance of deliberate, thoughtful approaches to development, modification, and deployment.

At a high level, AI systems serve as multifaceted tools capable of performing diverse tasks across multiple domains. Their general-purpose capabilities mean that a single system can support a variety of functions, from analytical processing to predictive modeling and knowledge synthesis. This flexibility offers organizations the opportunity to leverage AI in multiple areas simultaneously, enhancing productivity and providing a foundation for strategic innovation. Conceptually, the significance of these systems lies not merely in their technical sophistication, but in their ability to amplify human and organizational capabilities, support complex problem-solving, and facilitate scalable operational improvements. By providing a unified platform

for multiple functions, AI systems enable organizations to consolidate workflows, reduce redundancy, and adapt more readily to evolving operational demands.

Despite these potential benefits, the adaptation and modification of AI systems introduces a range of challenges and considerations that organizations must address thoughtfully. Modifications, whether operational, technical, or functional, can influence system behavior in ways that are not always immediately apparent. Even seemingly minor changes may affect output consistency, performance reliability, or the broader interactions of the system within operational ecosystems. These uncertainties underscore the importance of recognizing the potential risks inherent in adapting AI systems. Risks may include unintended outputs, decreased reliability under certain conditions, or emergent behaviors that were not anticipated during the initial design or deployment. By acknowledging the limitations and uncertainties associated with modifications, organizations can better anticipate areas that require monitoring, assessment, and mitigation. Thoughtful consideration of these challenges is essential to ensuring that AI systems continue to function effectively, consistently, and safely in dynamic environments.

Another central consideration relates to the broader operational and organizational impact of modifications. AI systems do not exist in isolation; they interact with workflows, decision-making processes, and human operators. Changes to system parameters, integration with external data sources, or adaptation for new operational contexts may produce ripple effects that extend beyond immediate outputs. This interconnectedness highlights the importance of evaluating

modifications in relation to their systemic impact, including potential indirect effects on decision-making processes, stakeholder interactions, and operational outcomes. By examining AI modifications from a holistic perspective, organizations can identify unintended consequences, manage dependencies, and anticipate emergent behaviors, thereby promoting more reliable and effective system performance.

In addressing these challenges, the adoption of best practices and guiding principles plays a critical role in supporting safe, effective, and responsible AI utilization. General approaches to risk mitigation, monitoring, and operational oversight can provide organizations with structured frameworks for managing both anticipated and unanticipated outcomes. Principles such as comprehensive documentation, systematic evaluation of modifications, and structured monitoring of system outputs contribute to greater transparency, accountability, and operational consistency. Similarly, the establishment of governance frameworks that clearly delineate roles, responsibilities, and decision-making authority supports coordinated management of AI systems and helps ensure that modifications are executed in alignment with organizational objectives. By integrating these foundational principles into operational practices, organizations can enhance reliability, reduce uncertainty, and support the sustainable use of AI capabilities.

Furthermore, continuous evaluation and assessment of AI systems is essential for maintaining system integrity and effectiveness. Ongoing monitoring enables organizations to detect deviations in output, identify performance changes, and address emergent challenges proactively. This iterative approach allows for the refinement of operational practices and the alignment of system behavior with intended objectives. Conceptually, continuous assessment reinforces both operational control and strategic oversight,

creating feedback loops that support adaptive improvement while mitigating potential risks associated with modifications. By embedding these practices into organizational workflows, organizations can maintain confidence in system performance while preserving the flexibility that makes AI systems valuable in the first place.

Acknowledging the progress that has been made in AI development and deployment is equally important. Research, experimentation, and iterative development have yielded insights into the behavior, limitations, and potential of AI systems. These advances have enhanced understanding of AI system dynamics, the implications of modifications, and methods for risk assessment and operational governance. The accumulation of conceptual and practical knowledge continues to inform best practices, refine modification strategies, and support more reliable deployment. While challenges and uncertainties remain, the trajectory of progress demonstrates that careful study, structured management, and systematic evaluation can improve the robustness and utility of AI systems over time. Organizations can leverage this ongoing progress to develop more effective management strategies, optimize operational outcomes, and support informed decision-making.

Looking forward, the future of AI system integration and modification will likely be shaped by continued technological advancement, deeper understanding of system behavior, and evolving operational requirements. Organizations can anticipate that AI systems will become increasingly sophisticated, capable of more complex interactions, and adaptable across a wider range of tasks. Consequently, the principles of thoughtful modification, systematic monitoring, and structured governance will remain central to maintaining operational reliability and effectiveness. Conceptual foresight suggests that organizations that adopt proactive, structured, and adaptive

approaches will be better positioned to manage uncertainties, capitalize on opportunities, and navigate the complex dynamics associated with AI system deployment.

From a strategic perspective, forward-looking organizations should consider AI modifications and management not as isolated technical exercises but as integral components of broader operational, organizational, and ethical considerations. Effective planning encompasses risk anticipation, governance alignment, and continuous assessment, supporting both short-term operational objectives and long-term organizational resilience. Furthermore, embedding adaptive principles into AI practices enables organizations to respond effectively to emergent behaviors, evolving operational needs, and shifts in stakeholder expectations. By maintaining a focus on conceptual integrity, operational reliability, and system adaptability, organizations can maximize the value of AI systems while mitigating potential challenges.

In conclusion, the integration and adaptation of AI systems constitutes a critical area of organizational focus, with significant potential to enhance operational capability, decision-making, and strategic performance. The benefits of AI systems stem from their flexibility, general-purpose capabilities, and capacity to process complex information across multiple domains. However, modifications to AI systems introduce considerations related to system behavior, operational dependencies, and potential risks. Thoughtful attention to these factors is essential to ensuring reliable, effective, and responsible use.

The adoption of guiding principles—including governance frameworks, structured documentation, risk assessment, and continuous evaluation—supports both operational consistency and accountability. These approaches provide conceptual and analytical tools for managing uncertainties,

monitoring system performance, and mitigating potential unintended outcomes. Ongoing research, iterative development, and the accumulation of practical insights continue to improve understanding of AI system behavior, modification impacts, and best practices.

Looking ahead, organizations should embrace adaptive, forward-looking approaches that balance operational optimization, innovation, and risk management. By maintaining focus on careful planning, responsible modification, and continuous oversight, organizations can harness the capabilities of AI systems effectively while minimizing uncertainties. The forward-looking perspective emphasizes that AI systems are dynamic, evolving entities whose value and reliability are closely tied to the quality of governance, evaluation, and operational stewardship. Conceptually, this underscores the importance of systematic, principle-driven management in achieving sustainable, high-value outcomes from AI systems, providing a foundation for continued progress and strategic advantage.

Ultimately, the careful integration and thoughtful modification of AI systems can enable organizations to realize substantial benefits while addressing challenges in a controlled and principled manner. By emphasizing structured governance, risk awareness, and adaptive oversight, organizations position themselves to navigate uncertainties responsibly, optimize operational performance, and support sustainable, effective utilization of AI capabilities. This conclusion highlights the need for continued attention to foundational principles, ongoing evaluation, and forward-looking planning as essential elements for managing the evolving landscape of AI systems effectively and responsibly.

Appendix: “Beyond Rules: When AI Modifies Itself”

The EU AI Act treats AI modification as a human-led process. But what happens when AI modifies itself?

We are entering the age of agentic AI—systems that self-prompt, self-tune, and act autonomously. Here, the regulator's "Who modified it?" becomes a profound, even philosophical, question.

The answer may not lie in more rules, but in better infrastructure. The emerging blueprint of Digital Public Infrastructure (DPI)—and directives like PSD3—shows how trust can be engineered into systems at scale. Imagine agentic AI that works within clear and responsible systems: its identity confirmed, its actions following the rules by design, and every change it makes recorded securely on a public ledger.

This concept is contextual integrity. Protocols like Model Context Protocol (MCP) are crucial systems that enable AI to clearly state its purpose, origin, and any changes in a way that machines can understand and trust.

The future belongs not to those who merely govern AI, but to those who build the world it can safely inhabit. The next regulatory frontier isn't a document—it's an Ecosystem.



Andrea Frosinini,
Business Development
Manager, XDC Foundation

Andrea is a strategic leader modernizing global trade finance. He forges key partnerships and champions digital ecosystems, leveraging emerging technologies to automate processes and build resilient, trust-driven frameworks for cross-border.

References

DAC Beachcroft, (2025), EU AI Act: Full Steam Ahead as GPAI Model Guidelines Published, accessible at: <https://www.dacbeachcroft.com/en/What-we-think/EU-AI-act-full-steam-ahead-as-GPAI-model-guidelines-published-and-rules-take-effect> (last accessed 26th January 2026)

EU Artificial Intelligence Act, (2025), Modifying AI Under the EU AI Act: Lessons from Practice on Classification and Compliance, accessible at: <https://artificialintelligenceact.eu/modifying-ai-under-the-eu-ai-act/> (last accessed 2nd May 2025)

European Commission AI Act Service Desk, (2024), Article 25: Responsibilities Along the AI Value Chain, accessible at: <https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-25> (last accessed 26th January 2026)

European Parliament and The Council of the European Union, (2024), 2024/1689 Regulation (EU) 2024/1689 of the European Parliament and of The Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act), accessible at https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_202401689 (last accessed 29th March 2025)

Härting, (2025), Provider or Deployer? Decoding the Key Roles in the AI Act, accessible at: <https://haerting.de/en/insights/provider-or-deployer-decoding-the-key-roles-in-the-ai-act/> (last accessed 26th January 2026)

ITMAGINATION, (2025), Making AI Work Under the EU AI Act: Practical Steps and Proven Patterns, accessible at: <https://www.itmagination.com/blog/making-ai-work-under-the-eu-ai-act-practical-steps-and-proven-patterns> (last accessed 26th January 2026)

Modulos, (2024), EU AI Act Compliance and Conformity Assessments, accessible at: <https://www.modulos.ai/eu-ai-act/> (last accessed 26th January 2026)

Naaia, (2025), Post-Market Monitoring: An Important AI Act Requirement, accessible at: <https://naaia.ai/post-market-monitoring-an-important-ai-act-requirement/> (last accessed 26th January 2026)

Portamedica, (2024), AI Act – Significant Change or Substantial Modification?, accessible at: <https://www.portamedica.eu/post/ai-act-significant-change-or-substantial-modification> (last accessed 26th January 2026)

Walker Morris, (2025), EU AI Act: An Introduction to High-Risk AI Systems, accessible at: <https://www.walkermorris.co.uk/comment-opinion/eu-ai-act-an-introduction-to-high-risk-ai-systems/> (last accessed 26th January 2026)



Important notice

This document has been prepared by AI & Partners B.V. for the sole purpose of enabling the parties to whom it is addressed to evaluate the capabilities of AI & Partners B.V. to supply the proposed services.

Other than as stated below, this document and its contents are confidential and prepared solely for your information, and may not be reproduced, redistributed or passed on to any other person in whole or in part. If this document contains details of an arrangement that could result in a tax or National Insurance saving, no such conditions of confidentiality apply to the details of that arrangement (for example, for the purpose of discussion with tax authorities). No other party is entitled to rely on this document for any purpose whatsoever and we accept no liability to any other party who is shown or obtains access to this document.

This document is not an offer and is not intended to be contractually binding. Should this proposal be acceptable to you, and following the conclusion of our internal acceptance procedures, we would be pleased to discuss terms and conditions with you prior to our appointment. Images used throughout the document have either been produced in-house or sourced from publicly available sources (see **References** for details).

AI & Partners B.V. is the Dutch headquarters of AI & Partners, a global professional services firm. Please see <https://www.ai-and-partners.com/> to learn more about us.

© 2025 AI & Partners B.V. All rights reserved.

Designed and produced by AI & Partners B.V.